

# LAB EXPERIMENT

## ITA0506-COMPUTER VISION

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**EXPNO:12**

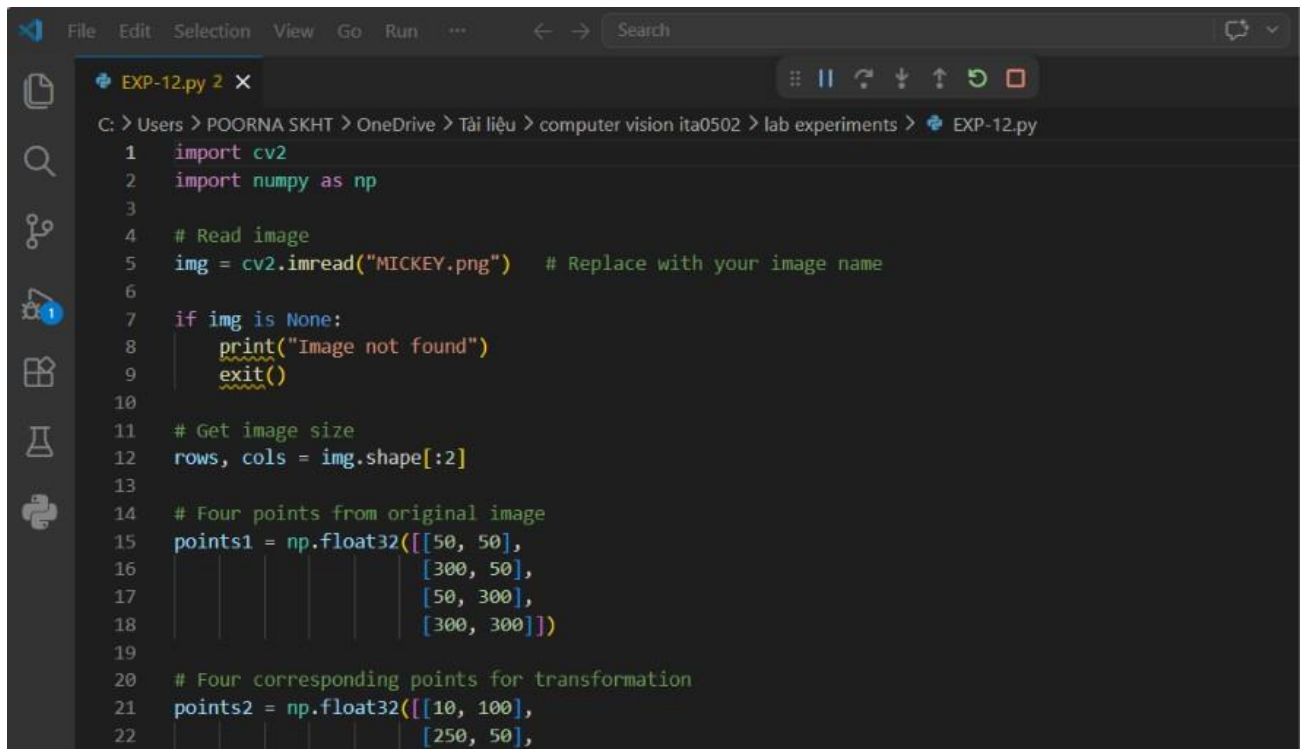
**Aim:**

To perform perspective transformation on an image using Python and OpenCV.

**Algorithm:**

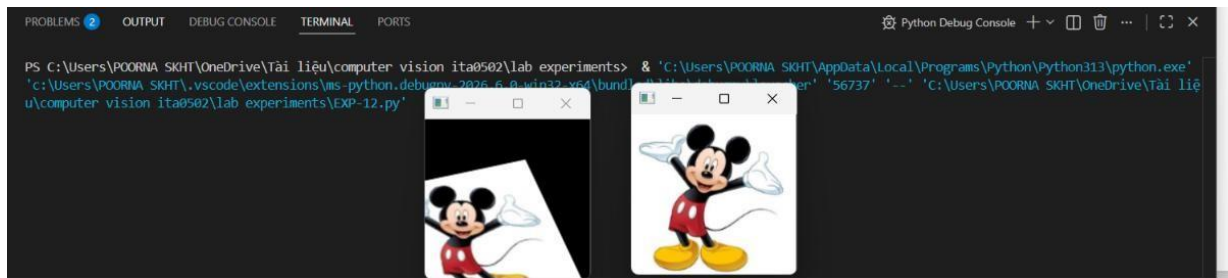
1. Import the required libraries (cv2 and numpy).
2. Read the input image.
3. Select four points from the original image.
4. Define the corresponding four points in the output image.
5. Compute the perspective transformation matrix using cv2.getPerspectiveTransform().
6. Apply the transformation using cv2.warpPerspective().
7. Display the original and transformed images.
8. Close all windows after a key press.

**CODE:**

A screenshot of a code editor window with a dark theme. The title bar shows 'EXP-12.py 2' and various icons. The file path is 'C:\> Users > POORNA SKHT > OneDrive > Tải liệu > computer vision ita0502 > lab experiments > EXP-12.py'. The code is as follows:

```
1  import cv2
2  import numpy as np
3
4  # Read image
5  img = cv2.imread("MICKEY.png") # Replace with your image name
6
7  if img is None:
8      print("Image not found")
9      exit()
10
11 # Get image size
12 rows, cols = img.shape[:2]
13
14 # Four points from original image
15 points1 = np.float32([[50, 50],
16                       [300, 50],
17                       [50, 300],
18                       [300, 300]])
19
20 # Four corresponding points for transformation
21 points2 = np.float32([[10, 100],
22                       [250, 50],
```

## OUTPUT :



**RESULT:** The perspective transformation was successfully performed on the image using Python and OpenCV.